

Karthik Rajgopal

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EDUCATION

University of California, Berkeley Aug 2024 - Present
MEng in Mechanical Engineering (Robotics & Control Systems track) [GPA: 3.58/4.0] California, USA
Courses: Experiential Advanced Control Design I (Model Predictive Control) & II (State Estimation), Modeling and Control of Multi-Agent Systems, Deep Tech Commercialization Strategies, Lean Launchpad

Birla Institute of Technology and Science, Pilani - Pilani Campus Aug 2019 – May 2023
Bachelor of Engineering, Mechanical Engineering [CGPA: 8.79/10] Rajasthan, India
Courses: Control Systems, Mechanisms & Machines, Computer-Aided Design, Advanced Manufacturing Process

SKILLS

Programming: Python, C, C++, MATLAB **CAD & FEA:** SolidWorks, Fusion 360, AutoCAD, Ansys
Robotics: ROS, Gazebo, PyBullet, Simulink **Others:** Arduino, 3D Printing, Figma

WORK EXPERIENCE

Project Assistant May 2023 – June 2024
Robert Bosch Centre for Cyber-Physical Systems, Indian Institute of Science Bengaluru, India

– **Collision Cone Control Barrier Functions (C3BF) [Publication]** | ROS, Python

- Worked on C3BF which combines the concepts of collision cone and control barrier functions (CBF) to provide a real-time safety filter (with mathematical guarantees) that can be implemented on top of a reference controller.
- Experimentally verified C3BF-QP-based controller on the Turtlebot (unicycle model) and a car (bicycle model).

– **Mechanical Design of Quadruped** | SolidWorks, ANSYS, PyBullet

- Designed a 3 DOF robotic leg driven using compound planetary actuators for a 25 kg payload capacity quadruped.
- Optimized knee joint performance through roller chain and bearing calculations for additional gear reduction.
- Simulated Static Structural Analysis using forces and moments data from MPC-based PyBullet simulations.

PROJECTS

Learning-to-win Drone Racing Sep 2024 – Present
Mentor: Prof. Claire Tomlin, Professor, UC Berkeley California, US

- Developing controllers based on deep reinforcement learning (RL) and game theory for competitive racing.
- Leveraging reachability analysis using max-min Deep Deterministic Policy Gradient (DDPG), and Multi-Agent Proximal Policy Optimization (MAPPO) to achieve strategic planning with safety and winning guarantees.

Quantum Circuits for Super Conducting Digital Computing Sep 2024 – Dec 2024
Mentor: Mr. Matthew Rappaport, UC Berkeley California, US

- Researched market needs and commercialization opportunities for superconducting digital computing.
- Interviewed 25+ industry leaders (IBM, NVIDIA, Intel, etc.) to assess market fit and industry challenges.
- Analyzed competitive and patent landscape to highlight unique differentiators and define strategic value.

Coverage Control for Aerial-Ground Robot Teams in Emergency Response Sep 2024 – Dec 2024
Mentor: Prof. Negar Mehr, Assistant Professor, UC Berkeley California, US

- Designed a hybrid Voronoi coverage controller for heterogeneous robots, reducing sensor loss time by 88%.
- Developed Python simulation environments using SciPy to optimize aerial-ground coordination.
- Enhanced sensor deployment speed and emergency response for chemical spill scenarios.

Optimized Path Planning for Parallel Parking | Python Sep 2024 – Dec 2024
Mentor: Dr. Charlott S. Vallon, UC Berkeley California, US

- Implemented a Model Predictive Control algorithm for precise and adaptive parallel parking path planning.
- Optimized algorithm parameters and cost functions, achieving final parking error <5 cm in simulation environments.

Design and RL-based Control of a Biped [Publication] | SolidWorks, MuJoCo Aug 2022 – April 2023
Mentor: Dr. Shishir NY Kolathaya, Assistant Professor, Indian Institute of Science Bengaluru, India

- Designed a low-cost, 3D-printed modular bipedal robot called StochBiRo which can traverse uneven terrains.
- Worked on developing a learning-based linear policy controller that handles unexpected terrain disturbances & external perturbations and also compensates for the low computational capabilities of the robot.

COMPETITIONS

Runners Up - Berkeley Venture Capital Investment Competition (VCIC) Dec 2024 - Feb 2025

- Conducted founder due diligence, valuation, and competitive analysis to assess venture investment opportunities.
- Participated in partner meetings, drafted term sheets with key terms, and performed market research on startups.